Grade 9 Mathematics

Exam study notes · November 2026
Paper 1: 2 hours, 100 marks · Paper 2: 90 minutes, 75 marks
Bring a calculator, ruler, pencil and eraser

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Paper 1 (2 hours, 100 marks)

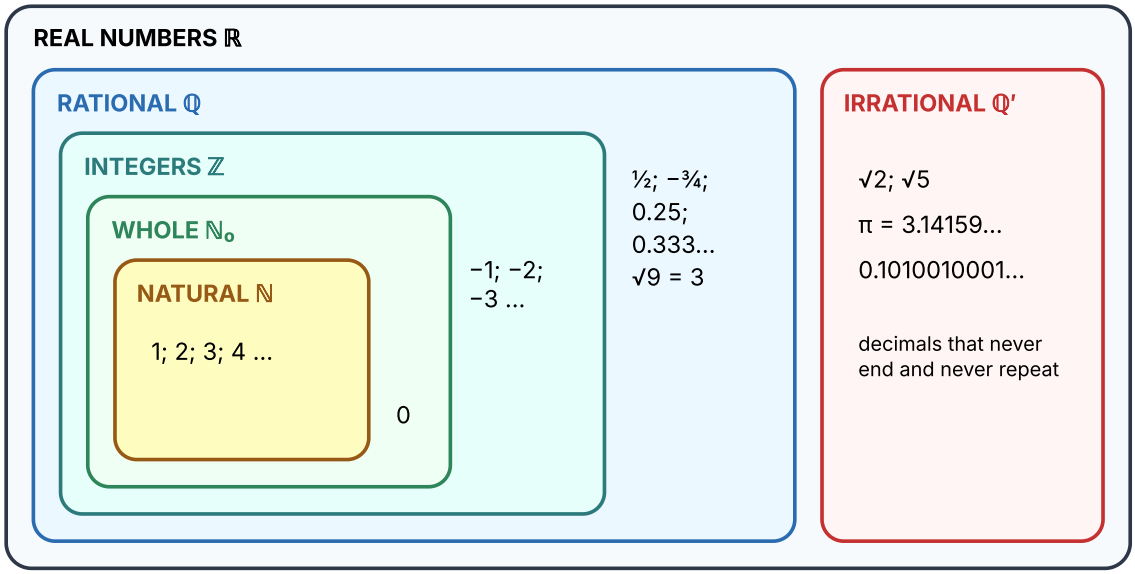
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Paper 2 (90 minutes, 75 marks)

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Paper 1 · 1. The real number system (chapter 1)

Numbers are sorted into **sets**. Each set fits inside a bigger one, like boxes inside boxes. Together they make up the **real numbers (ℝ)**.



The real number system: each set fits inside the next

Set	Symbol	What it contains	Examples
Natural numbers	$\mathbb{N}$	The counting numbers	1; 2; 3; 4; ...
Whole numbers	$\mathbb{N}_0$	Natural numbers and 0	0; 1; 2; 3; ...
Integers	$\mathbb{Z}$	Whole numbers and their negatives	...; −2; −1; 0; 1; 2; ...
Rational numbers	$\mathbb{Q}$	Numbers that **can be written as a fraction** a/b (b ≠ 0). As decimals they **end** or **repeat**	½; −3; 0.75; 0.333...; √16 = 4
Irrational numbers	$\mathbb{Q}'$	Numbers that **cannot** be written as a fraction. Their decimals **never end and never repeat**	√2; √3; √7; π
Real numbers	$\mathbb{R}$	All rational and irrational numbers together	every number on the number line

Watch out: √9 = 3 is **rational** (it's a whole number), but √10 is **irrational**. A square root is only rational if the number under it is a perfect square (1, 4, 9, 16, 25 ...). Also, 22/7 is rational, but π is irrational – 22/7 is only an approximation of π.

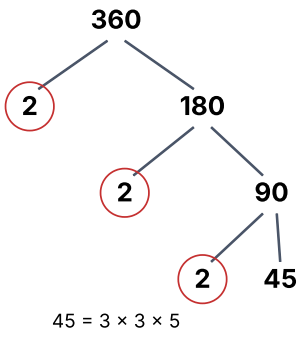
Properties of numbers

Property	Rule	Example
Commutative (order does not matter)	a + b = b + a; a × b = b × a	3 + 5 = 5 + 3; 4 × 7 = 7 × 4
Associative (grouping does not matter)	(a + b) + c = a + (b + c)	(2 + 3) + 4 = 2 + (3 + 4)
Distributive	a(b + c) = ab + ac	3(4 + 5) = 3 × 4 + 3 × 5 = 27
Additive identity	a + 0 = a	8 + 0 = 8
Multiplicative identity	a × 1 = a	8 × 1 = 8
Additive inverse	a + (−a) = 0	5 + (−5) = 0
Multiplicative inverse (reciprocal)	a × 1/a = 1	4 × ¼ = 1

Subtraction and division are **not** commutative: 7 − 3 ≠ 3 − 7. Division by **0** is **undefined** (not allowed).

Factors, multiples and primes

- **Factor:** a number that divides exactly into another. Factors of 12: 1; 2; 3; 4; 6; 12.
- **Multiple:** the answer when you multiply a number by 1, 2, 3 … Multiples of 4: 4; 8; 12; 16; …
- **Prime number:** has exactly **two** factors, 1 and itself: 2; 3; 5; 7; 11; 13; 17; 19; 23; 29 … (**1 is not prime**; **2 is the only even prime**).
- **Composite number:** has more than two factors: 4; 6; 8; 9; 10 …



Factor tree: 360 = $2^3 \times 3^2 \times 5$

Prime factorisation writes a number as a product of primes. Keep dividing by primes (2, 3, 5, 7 …) until only primes are left.

HCF and LCM of 24 and 36

$24 = 2 \times 2 \times 2 \times 3 = 2^3 \times 3$ $36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$

HCF (highest common factor) = primes in **both**, lowest powers = $2^2 \times 3$ = **12**

LCM (lowest common multiple) = **all** primes, highest powers = $2^3 \times 3^2$ = **72**

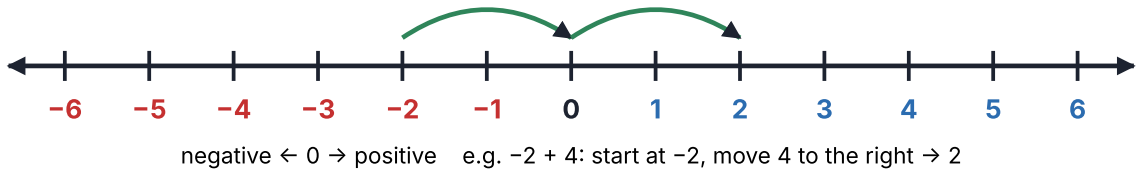
Ratio, rate and proportion

- A **ratio** compares quantities of the **same kind** (no units): boys to girls = 12 : 18 = **2 : 3** (simplify like a fraction).
- To **share in a ratio**: R500 shared in the ratio 2 : 3 → 5 parts; 1 part = R100 → **R200 and R300**.
- A **rate** compares **different** quantities (with units): 240 km in 3 hours = **80 km/h**.
- **Direct proportion:** both increase together. 3 pens cost R15 → 1 pen R5 → 7 pens **R35**.

- **Indirect (inverse) proportion:** one goes up, the other goes down. 4 workers take 6 days → 1 worker takes 24 days → 8 workers take **3 days**.

Paper 1 · 2. Integers (chapter 2)

Integers are whole numbers and their negatives: ...; −3; −2; −1; 0; 1; 2; 3; ... Numbers to the **right** on a number line are **bigger**.



The number line

Adding and subtracting integers

- Adding a **positive** number → move **right**. Adding a **negative** number → move **left**: 5 + (−3) = 5 − 3 = **2**.
- **Subtracting a negative is the same as adding**: 5 − (−3) = 5 + 3 = **8**.
- Same signs: add and keep the sign: −4 − 6 = **−10**. Different signs: subtract and take the sign of the bigger number: −9 + 4 = **−5**.

Multiplying and dividing integers

Signs	Answer	Example
+ × + or − × −	**positive**	(−4) × (−3) = 12
+ × − or − × +	**negative**	(−4) × 3 = −12
Division: same rules		(−20) ÷ (−5) = 4; 18 ÷ (−6) = −3

Same signs → positive. Different signs → negative. With many negatives: an **even** number of negatives gives a positive answer; an **odd** number gives a negative answer.

Powers and roots of integers

- $(-2)^2 = (-2) \times (-2) = \mathbf{4}$ but $-2^2 = -(2 \times 2) = \mathbf{-4}$ (the brackets matter!).
- $(-2)^3 = (-2) \times (-2) \times (-2) = \mathbf{-8}$. An even power of a negative is positive; an odd power is negative.
- $\sqrt[3]{(-27)} = \mathbf{-3}$, because $(-3)^3 = -27$. But $\sqrt{(-9)}$ is **not real** – no real number squared gives a negative.

Order of operations (BODMAS)

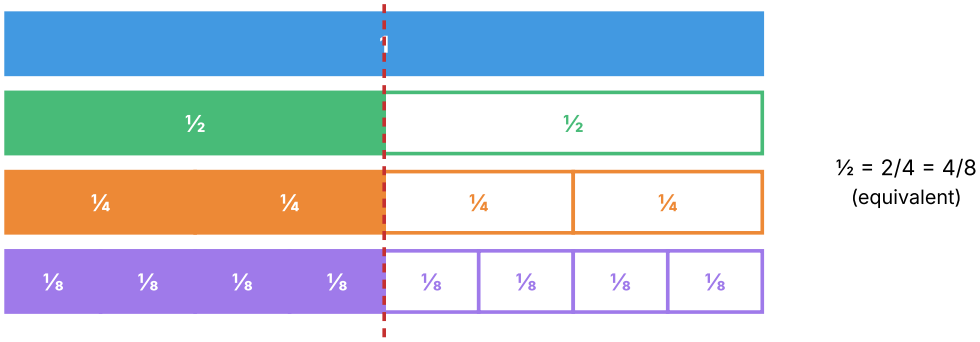
Brackets → **O**rders (powers and roots) → **D**ivision and **M**ultiplication (left to right) → **A**ddition and **S**ubtraction (left to right).

$3 + 4 \times 2^2 = 3 + 4 \times 4 = 3 + 16 =$ **19**
$-12 \div (-3) + 5 \times (-2) = 4 + (-10) =$ **−6**
$(-3)^2 - 2(4 - 7) = 9 - 2(-3) = 9 + 6 =$ **15**

Word problem: the temperature in Sutherland is −4 °C at 6 am and rises 11 °C by noon. Noon temperature = −4 + 11 = **7 °C**. Difference between 7 °C and −4 °C = 7 − (−4) = **11 °C**.

Paper 1 · 3. Common fractions (chapter 3)

A **fraction** shows part of a whole. In ¾, the **numerator** (3, on top) tells how many parts you have and the **denominator** (4, at the bottom) tells how many equal parts the whole is divided into.



Fraction wall: the shaded parts are all the same size – equivalent fractions

Types of fractions

- **Proper fraction:** numerator smaller than denominator: ⅗.
- **Improper fraction:** numerator bigger than denominator: 7/4.
- **Mixed number:** a whole number and a fraction: 1¾.
- **Mixed → improper:** 2⅓ = (2 × 3 + 1)/3 = **7/3**. **Improper → mixed:** 17/5 = 17 ÷ 5 = 3 remainder 2 = **3⅖**.
- **Equivalent fractions** have the same value: multiply or divide the top and bottom by the same number. 3/4 = 6/8 = 9/12.
- **Simplest form:** divide the top and bottom by the HCF: 18/24 = (18 ÷ 6)/(24 ÷ 6) = ¾.

Calculating with fractions

Operation	How	Example
Add / subtract	Make the denominators the same (use the LCD), then add or subtract the numerators only	⅔ + ¼ = 8/12 + 3/12 = **11/12** 2½ − 1⅔ = 5/2 − 5/3 = 15/6 − 10/6 = **5/6**
Multiply	Change mixed numbers to improper fractions; multiply top × top and bottom × bottom (simplify first if you can)	⅔ × ¾ = 6/12 = ½ 1½ × 2⅔ = 3/2 × 8/3 = 24/6 = **4**
Divide	**Keep, change, flip:** keep the first fraction, change ÷ to ×, flip the second fraction (reciprocal)	¾ ÷ ⅜ = ¾ × 8/3 = 24/12 = **2** 2 ÷ ⅕ = 2 × 5 = **10**
Fraction of an amount	"of" means ×: divide by the denominator, multiply by the numerator	⅗ of R200 = 200 ÷ 5 × 3 = **R120**
Squares and roots	Square or root the top and bottom separately	$(⅔)^2$ = **4/9**; √(9/25) = ⅗; $\sqrt[3]{(8/27)}$ = ⅔

> **Never add the denominators:** ½ + ⅓ is **not** 2/5. Find a common denominator: 3/6 + 2/6 = 5/6.

Fractions and percentages

- **Fraction → percentage:** × 100%. ⅗ = 3/5 × 100% = **60%**.
- **Percentage → fraction:** put it over 100 and simplify. 35% = 35/100 = **7/20**.
- **Percentage of an amount:** 15% of R480 = 15/100 × 480 = **R72**.
- **One amount as a percentage of another:** 18 out of 24 = 18/24 × 100% = **75%**.
- **Increase by a percentage:** R250 increased by 20% = 250 + 50 = **R300**. **Decrease:** R250 decreased by 20% = 250 − 50 = **R200**.

Paper 1 · 4. Decimal fractions (chapter 4)

A **decimal fraction** is another way of writing a fraction with a denominator of 10, 100, 1 000 … The **decimal comma/point** separates the whole number from the part.

thousands 1 000	hundreds 100	tens 10	units 1		tenths 1/10	hundredths 1/100	thousandths 1/1 000
3	4	5	2	,	6	7	8

3 452,678 → the 6 is worth 6/10, the 7 is worth 7/100, the 8 is worth 8/1 000

Place value of decimals

Converting between fractions, decimals and percentages

Change	How	Example
Fraction → decimal	Divide the numerator by the denominator (or make the denominator 10, 100, 1 000)	⅜ = 3 ÷ 8 = **0,375**; 7/20 = 35/100 = **0,35**
Decimal → fraction	Write it over 10, 100 or 1 000 and simplify	0,45 = 45/100 = **9/20**; 0,125 = 125/1 000 = **⅛**
Decimal → percentage	× 100	0,07 = **7%**; 1,2 = **120%**
Percentage → decimal	÷ 100	38% = **0,38**; 2,5% = **0,025**

- **Terminating decimals** end: ¼ = 0,25. **Recurring decimals** repeat forever: ⅓ = 0,333… = $0,\dot{3}$ (a dot or bar over the repeating digit); 2/11 = 0,1818… = $0,1\dot{8}$. Both are **rational**.
- **Comparing decimals:** line up the commas and compare digit by digit from the left: 0,45 > 0,405 (the tenths are the same; then 5 hundredths > 0 hundredths).

Calculating with decimals

Operation	How	Example
Add / subtract	Line up the decimal commas; fill gaps with 0	12,5 + 3,47 = 12,50 + 3,47 = **15,97**
× or ÷ by 10, 100, 1 000	× → move the comma **right** 1, 2, 3 places; ÷ → move it **left**	3,45 × 100 = **345**; 3,45 ÷ 100 = **0,0345**
Multiply	Multiply as whole numbers; the answer has as many decimal places as both numbers together	0,3 × 0,04: 3 × 4 = 12; 1 + 2 = 3 places → **0,012**
Divide by a decimal	Make the divisor a whole number by multiplying both numbers by 10, 100 …	4,8 ÷ 0,6 = 48 ÷ 6 = **8**
Squares and roots	Use the fraction or count decimal places	$0{,}3^2$ = **0,09**; √0,16 = **0,4**; $0{,}2^3$ = **0,008**

Rounding

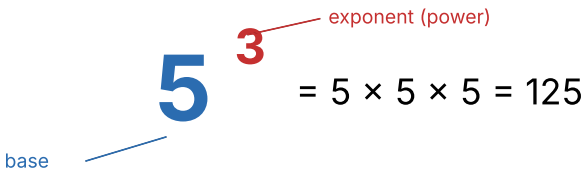
Look at the digit **after** the place you are rounding to: **5 or more → round up**; **4 or less → leave it**.

3,467 to 1 decimal place → **3,5** (the 6 rounds the 4 up)
3,467 to 2 decimal places → **3,47**; to the nearest whole number → **3**
Money is always rounded to 2 decimal places: R12,3456 → **R12,35**

South Africa uses a **decimal comma** (3,5) – your calculator shows a point (3.5). Both mean the same thing.

Paper 1 · 5. Exponents and scientific notation (chapter 5)

In $\mathbf{a^n}$, *a* is the **base** and *n* is the **exponent** (power). The exponent tells you how many times to multiply the base by itself: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = \mathbf{32}$.



"5 to the power 3" or "5 cubed"

The laws of exponents

Law	Rule	Example
Multiply, same base → **add** exponents	$a^m \times a^n = a^{m+n}$	$x^3 \times x^4 = \mathbf{x^7}$; $2^3 \times 2^2 = 2^5 = 32$
Divide, same base → **subtract** exponents	$a^m \div a^n = a^{m-n}$	$y^8 \div y^3 = \mathbf{y^5}$
Power of a power → **multiply** exponents	$(a^m)^n = a^{mn}$	$(x^2)^3 = \mathbf{x^6}$
Power of a product	$(ab)^n = a^n b^n$	$(2x)^3 = \mathbf{8x^3}$; $(3a^2b)^2 = \mathbf{9a^4b^2}$
Power of a quotient	$(a/b)^n = a^n/b^n$	$(x/3)^2 = \mathbf{x^2/9}$
Zero exponent	$a^0 = 1$ $(a \neq 0)$	$5^0 = 1$; $(3xy)^0 = \mathbf{1}$; $4x^0 = 4 \times 1 = \mathbf{4}$
Negative exponent	$a^{-n} = 1/a^n$	$2^{-3} = 1/2^3 = ⅛$; $x^{-2} = \mathbf{1/x^2}$

Simplify: work out the numbers and the variables separately.
$3x^2y \times 4x^5y^3 = 12x^7y^4$
$12x^6 \div 4x^2 = \mathbf{3x^4}$
$(2a^3)^4 = 2^4a^{12} = \mathbf{16a^{12}}$
$(-2)^4 = \mathbf{16}$ but $-2^4 = \mathbf{-16}$; $(-2)^3 = \mathbf{-8}$

Squares, cubes and roots

- $\sqrt{49} = 7$ because $7^2 = 49$. $\sqrt[3]{64} = 4$ because $4^3 = 64$. $\sqrt[3]{(-27)} = -3$.
- Learn the squares up to $15^2 = 225$ and the cubes up to $6^3 = 216$ (or $10^3 = 1\,000$).
- $\sqrt{(a \times b)} = \sqrt{a} \times \sqrt{b}$: $\sqrt{(16 \times 9)} = 4 \times 3 = 12$. But **$\sqrt{(9 + 16)} = \sqrt{25} = 5$, not $3 + 4$** – you cannot split a root over + or −.
- $\sqrt{(x^6)} = x^3$; $\sqrt[3]{(x^9)} = x^3$ (divide the exponent by 2 for a square root, by 3 for a cube root).

Scientific notation

Scientific notation is a short way to write very large or very small numbers: **$a \times 10^n$**, where **$1 \leq a < 10$** (one non-zero digit before the comma) and n is an integer.

Number	Move the comma	Scientific notation
45 000 000	7 places left → positive exponent	**$4{,}5 \times 10^7$**
384 400 (distance to the Moon in km)	5 places left	**$3{,}844 \times 10^5$**
0,00032	4 places right → negative exponent	**$3{,}2 \times 10^{-4}$**
0,000 000 7	7 places right	**7×10^{-7}**

Calculating in scientific notation

$(3 \times 10^4) \times (2 \times 10^3) = 6 \times 10^{4+3} =$ **6×10^7**

$(8 \times 10^9) \div (2 \times 10^3) = 4 \times 10^{9-3} =$ **4×10^6**

$(5 \times 10^3) \times (4 \times 10^2) = 20 \times 10^5 =$ **2×10^6** (adjust so that a is between 1 and 10)

Writing it out in full: $2{,}07 \times 10^4 =$ **20 700**; $5{,}1 \times 10^{-3} =$ **0,0051**

Paper 1 · 6. Number patterns (chapter 6)

A **number pattern** (sequence) is a list of numbers that follow a rule. Each number is a **term**: T_1 is the 1st term, T_2 the 2nd, ... T_n the n-th term (general term).

Linear patterns (constant difference)

If you **add or subtract the same number** each time, the pattern has a **constant difference**. The general term is $\mathbf{T_n = dn + c}$, where d is the constant difference.

Example: 5; 8; 11; 14; ...
Differences: 3; 3; 3 → constant difference d = 3, so $T_n = 3n + c$
Use $T_1 = 5$: $3(1) + c = 5 \rightarrow c = 2$. So $\mathbf{T_n = 3n + 2}$ (check: $T_2 = 3(2) + 2 = 8$ ✓)
50th term: $T_{50} = 3(50) + 2 =$ **152**
Which term is 89? $3n + 2 = 89 \rightarrow 3n = 87 \rightarrow n = 29 \rightarrow$ it is the **29th term**

Shortcut: $c = T_1 - d$ (the "zero term" – the number before the first term). For 5; 8; 11 ... the zero term is 5 − 3 = 2.

Patterns with pictures



Figure 1
4 sticks

Figure 2
7 sticks

Figure 3
10 sticks

Squares made from matches: 4; 7; 10; ... sticks

Number of sticks: 4; 7; 10 → constant difference 3 → $T_n = 3n + 1$
Figure 20 needs $3(20) + 1 =$ **61 sticks**. With 100 sticks: $3n + 1 = 100 \rightarrow n = 33 \rightarrow$ **figure 33** (uses exactly 100 sticks).

Other types of patterns

Type	Rule	Example
Constant ratio (geometric)	Multiply by the same number each time	3; 6; 12; 24; ... (× 2) → $T_n = 3 \times 2^{n-1}$ 80; 40; 20; 10; ... (× ½)
Neither (e.g. squares)	The difference changes, but the second difference may be constant	1; 4; 9; 16; ... → $T_n = n^2$ 2; 5; 10; 17; ... → $T_n = n^2 + 1$
Patterns in tables	Find the rule linking the position (n) and the value	n: 1, 2, 3, 4 → value: 7, 12, 17, 22 → $T_n = 5n + 2$

How to find the rule: (1) find the differences between terms; (2) if they are constant, use $T_n = dn + c$; (3) if not, check whether you multiply by the same number (ratio) or whether it is linked to n^2, n^3 ...; (4) always check your rule with two terms.

Paper 1 · 7. Algebraic expressions (chapter 8)

An **algebraic expression** uses letters (variables) to stand for numbers, e.g. **$3x^2 - 5x + 7$**. You must be able to name its parts, simplify it, multiply (expand) it and factorise it.

Words you must know

Word	Meaning	In $3x^2 - 5x + 7$
Term	Parts separated by + or − signs	3 terms: $3x^2$, $-5x$, 7
Coefficient	The number in front of a variable	3 (of x^2) and −5 (of x)
Constant	A term with no variable	7
Degree	The highest exponent	2
Monomial / binomial / trinomial	1 term / 2 terms / 3 terms	a trinomial
Like terms	Same variables with the same exponents	$3x^2$ and $-7x^2$ are like terms; $3x^2$ and 3x are not

Adding and subtracting: collect like terms

Only like terms can be added or subtracted. Add the coefficients and keep the variable part the same.

$4a + 3b - a + 2b = (4a - a) + (3b + 2b) =$ **$3a + 5b$**
$5x^2 - 2x + 3x^2 + 7x - 4 =$ **$8x^2 + 5x - 4$**

Multiplying

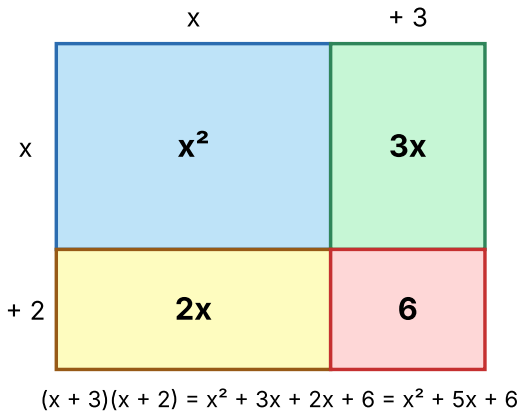
Multiply the **numbers** and add the **exponents** of the same variable: $3x^2y \times 4x^5y^3 = 12x^7y^4$.

Monomial × binomial: multiply every term inside the brackets

$3x(2x - 5) = 3x \times 2x + 3x \times (-5) =$ **$6x^2 - 15x$**
$-2a(a^2 + 4a - 1) =$ **$-2a^3 - 8a^2 + 2a$** (a minus outside changes every sign)

Binomial × binomial

Multiply **each term in the first bracket by each term in the second**. The area diagram shows why:



The four parts of the rectangle are the four products

$(x + 3)(x - 5) = x^2 - 5x + 3x - 15 = \mathbf{x^2 - 2x - 15}$
$(2x - 1)(x + 4) = 2x^2 + 8x - x - 4 = \mathbf{2x^2 + 7x - 4}$

Special products

Product	Rule	Example
Square of a sum	$(a + b)^2 = a^2 + 2ab + b^2$	$(x + 4)^2 = x^2 + 8x + 16$
Square of a difference	$(a - b)^2 = a^2 - 2ab + b^2$	$(2x - 3)^2 = 4x^2 - 12x + 9$
Difference of two squares	$(a + b)(a - b) = a^2 - b^2$	$(x + 7)(x - 7) = x^2 - 49$

Common mistake: $(x + 4)^2$ is **not** $x^2 + 16$. Write it as $(x + 4)(x + 4)$ and multiply out – the middle term $8x$ is easy to forget.

Dividing

Divide **every** term by the monomial: $(12x^3 - 8x^2 + 4x) \div 4x = \mathbf{3x^2 - 2x + 1}$.

Factorising (the opposite of expanding)

To **factorise** means to write an expression as a **product** (things multiplied). Always look for a **common factor first**.

Type	How	Examples
Common factor	Take out the highest number and variable that divide into every term	$6x^2 + 9x = 3x(2x + 3)$ $5a^2b - 10ab^2 = 5ab(a - 2b)$
Difference of two squares	$a^2 - b^2 = (a + b)(a - b)$. Both terms must be perfect squares with a minus between them	$x^2 - 25 = (x + 5)(x - 5)$ $9y^2 - 16 = (3y + 4)(3y - 4)$ $2x^2 - 50 = 2(x^2 - 25) = 2(x + 5)(x - 5)$
Trinomial $x^2 + bx + c$	Find two numbers that **multiply to c** and **add to b**	$x^2 + 7x + 12 = (x + 3)(x + 4)$ [3 × 4 = 12, 3 + 4 = 7] $x^2 - x - 12 = (x - 4)(x + 3)$ [−4 × 3 = −12, −4 + 3 = −1] $x^2 - 8x + 15 = (x - 3)(x - 5)$

Check: you can always check a factorisation by multiplying it out again – you should get the original expression.

Simplifying algebraic fractions

Factorise the top and bottom first, then cancel **common factors** (never cancel single terms).

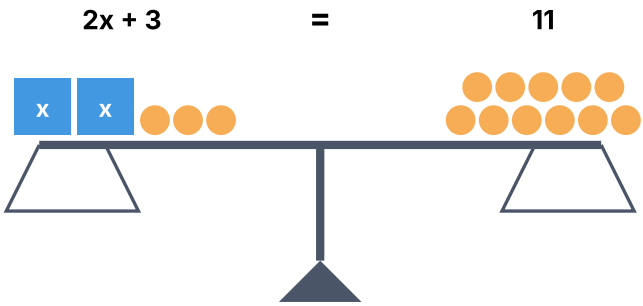
$(x^2 - 9) \div (x + 3) = (x + 3)(x - 3) \div (x + 3) =$ **$x - 3$**
$(4x + 8) \div 4 = 4(x + 2) \div 4 =$ **$x + 2$**

Substitution

Replace the variable with the number, using brackets: if $x = -2$, then $3x^2 - x = 3(-2)^2 - (-2) = 3(4) + 2 =$ **14**.

Paper 1 · 8. Algebraic equations (chapter 9)

An **equation** is a number sentence with an equal sign. To **solve** it, find the value of the variable that makes both sides equal. An equation is like a **balanced scale**: whatever you do to one side, you must do to the other.



2x + 3 = 11: take 3 off both sides → 2x = 8; divide both sides by 2 → x = 4

Steps for solving

1. Remove **brackets** (multiply out).
2. Remove **fractions** by multiplying every term by the LCD (lowest common denominator).
3. Get all the **variable terms on one side** and the numbers on the other (do the opposite operation to both sides).
4. **Divide** by the coefficient of the variable.
5. **Check** by substituting your answer back into the original equation.

Type	Example	Solution
Simple linear	$3x - 7 = 11$	$3x = 18 \rightarrow$ **$x = 6$**
Variable on both sides	$5x + 2 = 2x - 10$	$3x = -12 \rightarrow$ **$x = -4$**
With brackets	$2(x - 3) = x + 4$	$2x - 6 = x + 4 \rightarrow$ **$x = 10$**
With fractions	$x/3 + x/2 = 5$	$\times 6$: $2x + 3x = 30 \rightarrow 5x = 30 \rightarrow$ **$x = 6$**
Product equals zero	$(x - 2)(x + 5) = 0$	$x - 2 = 0$ or $x + 5 = 0 \rightarrow$ **$x = 2$ or $x = -5$**
Factorise first	$x^2 - 5x + 6 = 0$	$(x - 2)(x - 3) = 0 \rightarrow$ **$x = 2$ or $x = 3$**
Common factor	$x^2 - 4x = 0$	$x(x - 4) = 0 \rightarrow$ **$x = 0$ or $x = 4$**
Exponential	$2^x = 32$	$2^x = 2^5 \rightarrow$ **$x = 5$**

Why "product = 0" works: if two numbers multiply to 0, at least one of them must be 0. So set **each bracket equal to 0**. Make sure one side is 0 **before** you factorise.

Checking: $5x + 2 = 2x - 10$ with $x = -4$: LHS $= 5(-4) + 2 = -18$; RHS $= 2(-4) - 10 = -18$. LHS = RHS ✓

Word problems

1. Let x = the unknown (write it down).
2. Turn the words into an equation.
3. Solve, then answer the question in words.

Example 1: The sum of three consecutive numbers is 72. Find the numbers.
Let the numbers be x, x + 1 and x + 2 → x + (x + 1) + (x + 2) = 72
3x + 3 = 72 → 3x = 69 → x = 23. The numbers are **23, 24 and 25**.

Example 2: Thabo is 3 times as old as his sister. In 4 years the sum of their ages will be 32. How old are they now?
Sister = x, Thabo = 3x → (x + 4) + (3x + 4) = 32
4x + 8 = 32 → 4x = 24 → x = 6. Sister is **6**, Thabo is **18**.

Example 3: The length of a rectangle is 5 cm more than its breadth. The perimeter is 38 cm. Find its dimensions.
Breadth = x, length = x + 5 → 2x + 2(x + 5) = 38
4x + 10 = 38 → 4x = 28 → x = 7. Breadth = **7 cm**, length = **12 cm**.

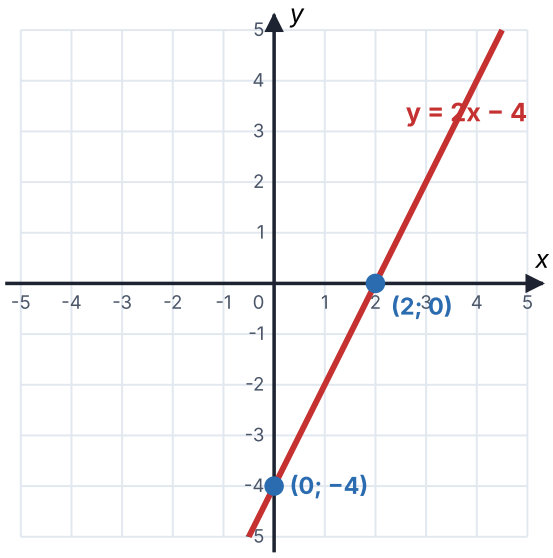
Paper 1 · 9. Graphs (chapter 16)

A graph shows the relationship between two variables. On the **Cartesian plane**, the horizontal axis is the **x-axis** and the vertical axis is the **y-axis**. They cross at the **origin (0; 0)**. A point is written as **(x; y)** – x first, then y.

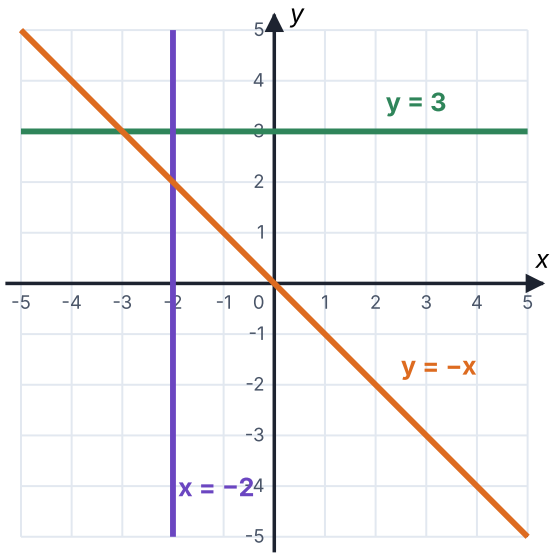
Linear graphs: y = mx + c

A **linear function** makes a **straight line**. In **y = mx + c**:

- **m = the gradient** (slope): how steep the line is. Gradient = rise ÷ run = $(y_2 - y_1) \div (x_2 - x_1)$.
  - m positive → the line goes **up** to the right; m negative → **down** to the right; m = 0 → **horizontal**.
- **c = the y-intercept**: where the line cuts the y-axis (put x = 0).
- The **x-intercept** is where the line cuts the x-axis (put y = 0 and solve).



y = 2x − 4: gradient 2, y-intercept −4, x-intercept 2



Special lines: horizontal, vertical, and through the origin

How to draw a straight-line graph

Method 1 – dual intercepts: draw y = 2x − 4
y-intercept: put x = 0 → y = −4 → point (0; −4)
x-intercept: put y = 0 → 0 = 2x − 4 → x = 2 → point (2; 0)
Plot both points and draw a straight line through them with a ruler. Label the line with its equation.

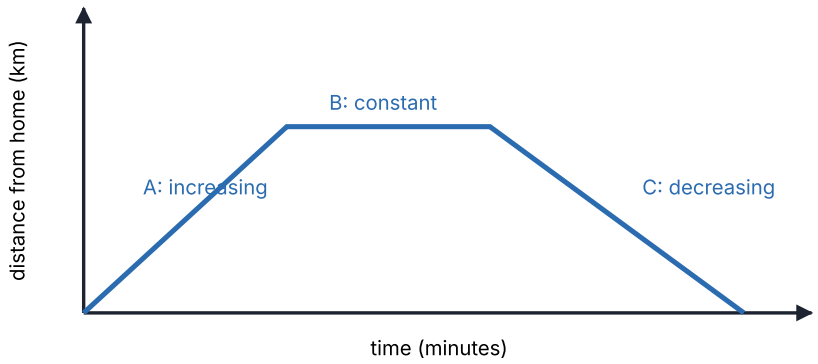
Method 2 – table of values: choose x-values, work out y, plot the points.
y = 2x − 4: x = −1 → y = −6; x = 0 → y = −4; x = 1 → y = −2; x = 2 → y = 0; x = 3 → y = 2

Gradient from two points: the line through (1; 3) and (4; 9)
m = (9 − 3) ÷ (4 − 1) = 6 ÷ 3 = **2**

Equation	What the line looks like
y = k (e.g. y = 3)	**Horizontal** line through 3 on the y-axis; gradient = 0
x = k (e.g. x = −2)	**Vertical** line through −2 on the x-axis; gradient is undefined
y = mx (no c)	Goes through the **origin** (0; 0)
Same m, e.g. y = 3x + 1 and y = 3x − 5	**Parallel** lines (same steepness)

Where two lines meet (point of intersection): y = x + 1 and y = −x + 5
Set them equal: x + 1 = −x + 5 → 2x = 4 → x = 2; then y = 2 + 1 = 3 → **(2; 3)**

Reading and interpreting graphs



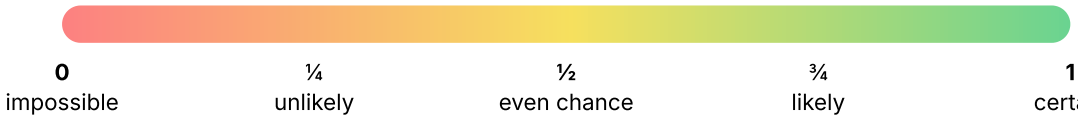
A learner walks away from home (A), stops at a shop (B) and walks back home (C)

Word	Meaning
Increasing	The graph goes up from left to right
Decreasing	The graph goes down from left to right
Constant	The graph is flat – no change (on a distance–time graph the person has stopped)
Linear / non-linear	A straight line / a curve
Discrete / continuous	Separate points that are not joined (e.g. number of people) / a joined line (e.g. time, temperature)
Maximum / minimum	The highest / lowest point of the graph

On a distance–time graph, a **steeper** line means **faster** movement.

Paper 1 · 10. Probability (chapter 20)

Probability measures how likely something is to happen. It is a number from **0 (impossible)** to **1 (certain)**, written as a fraction, decimal or percentage.



The probability scale

P(event) = number of favourable outcomes ÷ total number of possible outcomes

Word	Meaning
Outcome	One possible result (rolling a 4)
Event	The outcome(s) you are interested in (rolling an even number)
Sample space	A list of **all** possible outcomes. Dice: {1; 2; 3; 4; 5; 6}
Complement	The event **not** happening: P(not A) = 1 − P(A)

One die:
P(even) = 3 ÷ 6 = ½ (2, 4, 6)
P(number greater than 4) = 2 ÷ 6 = ⅓ (5, 6)
P(not a 6) = 1 − ⅙ = ⅚ P(7) = **0** (impossible)

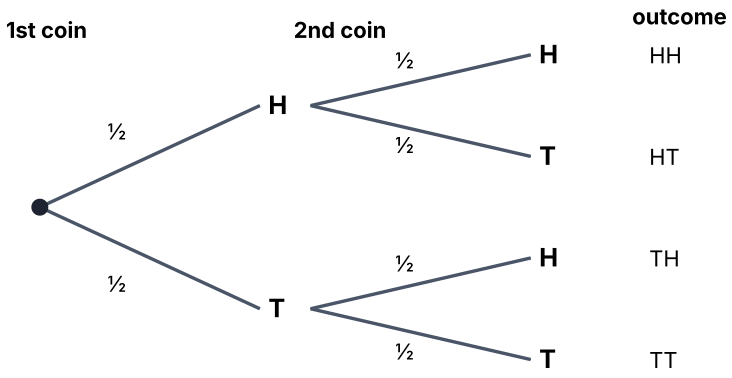
A bag has 5 red, 3 blue and 2 green beads (10 beads).
P(red) = 5 ÷ 10 = ½ P(blue) = **3/10** P(not green) = 8 ÷ 10 = ⅘

Two events: tables and tree diagrams

When two things happen together (e.g. a coin **and** a die), list all the outcomes in a **table** or a **tree diagram**. Total outcomes = outcomes of the first × outcomes of the second.

	1	2	3	4	5	6
H	H1	H2	H3	H4	H5	H6
T	T1	T2	T3	T4	T5	T6

Coin and die: 2 × 6 = 12 outcomes. P(heads and a 6) = 1/12. P(tails and an even number) = 3/12 = ¼.



Tree diagram for tossing two coins: 4 outcomes. P(two heads) = ¼; P(one head and one tail) = 2/4 = ½.

Theoretical probability and relative frequency

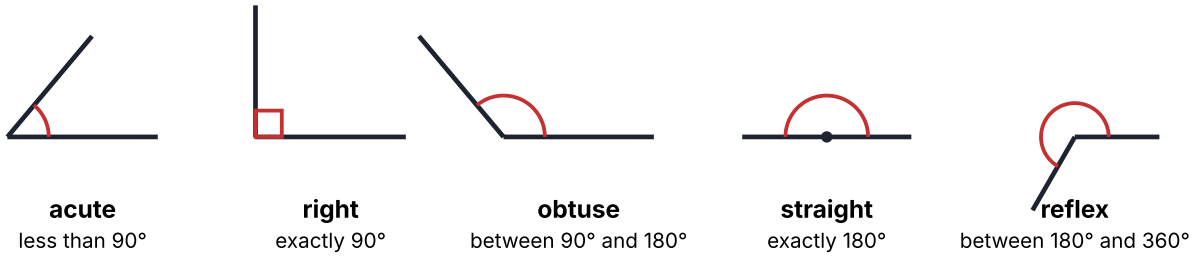
- **Theoretical probability:** what **should** happen, worked out with the formula. P(heads) = ½.
- **Relative frequency (experimental probability):** what **actually** happened in an experiment = number of times the event happened ÷ number of trials.
- The **more trials** you do, the **closer** the relative frequency gets to the theoretical probability.

Example: A coin is tossed 50 times and lands on heads 30 times.
Relative frequency of heads = 30 ÷ 50 = **0.6** (theoretical probability = 0.5)

Paper 2 · 11. Geometry of straight lines (chapter 11)

In geometry you must give a **reason** for every statement, in brackets. Learn the facts **and** the short reasons.

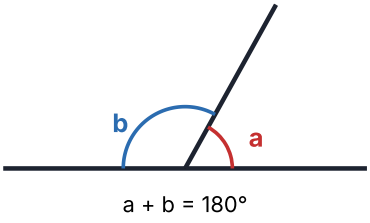
Types of angles



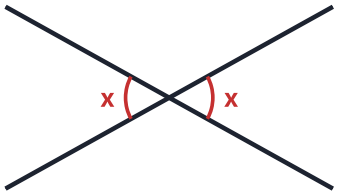
Types of angles

Also: a **revolution** is 360°. **Complementary** angles add up to **90°**; **supplementary** angles add up to **180°**.

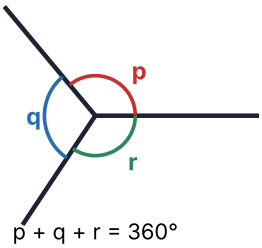
Angle facts on straight lines



∠s on a straight line add up to 180° (∠s on a str line)



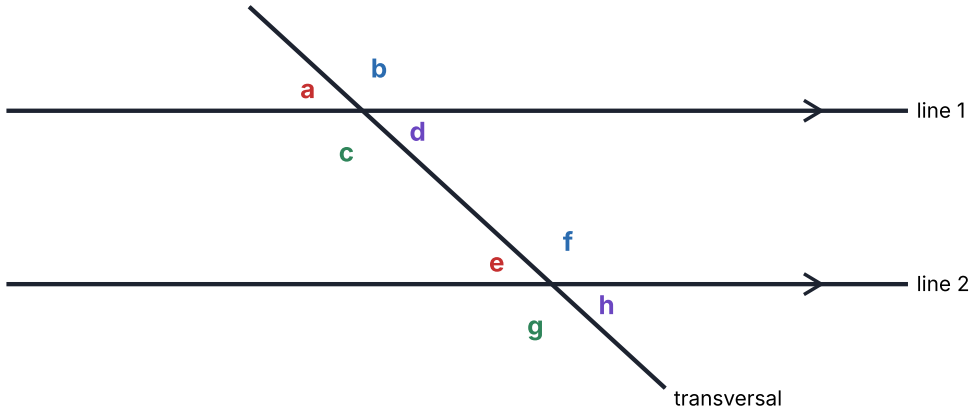
Vertically opposite ∠s are equal (vert opp ∠s)



∠s around a point add up to 360° (∠s round a pt)

Parallel lines cut by a transversal

A **transversal** is a line that cuts across two or more lines. When it cuts **parallel lines**, special pairs of angles are formed.



Line 1 ∥ line 2 (the arrows show the lines are parallel)

Angle pair	Shape to look for	Rule	In the diagram	Reason
Corresponding angles	**F**-shape (same position at each line)	Equal	a = e, b = f, c = g, d = h	(corr ∠s; line 1 ∥ line 2)
Alternate angles	**Z**-shape (inside the lines, opposite sides of the transversal)	Equal	c = f, d = e	(alt ∠s; line 1 ∥ line 2)
Co-interior angles	**U** / C-shape (inside, same side)	Add up to 180°	c + e = 180°, d + f = 180°	(co-int ∠s; line 1 ∥ line 2)

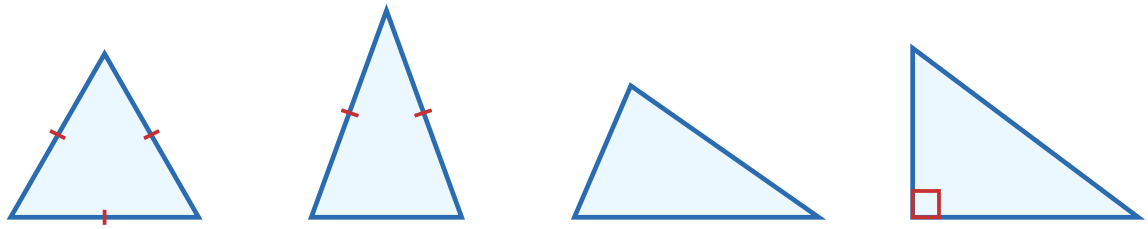
To prove lines are parallel, use the rules the other way round: if corresponding angles are equal, or alternate angles are equal, or co-interior angles add up to 180°, then the lines are parallel.

Example: In the diagram, a = 70°. Find all the other angles.
b = 180° − 70° = **110°** (∠s on a str line); d = a = **70°** (vert opp ∠s); c = b = **110°** (vert opp ∠s)
e = a = **70°** (corr ∠s); f = b = **110°** (corr ∠s); g = c = **110°**; h = d = **70°**

Other words: **perpendicular** lines meet at 90° (symbol ⊥); **parallel** lines never meet (symbol ∥); a **line segment** has two end points; a **ray** has one end point.

Paper 2 · 12. Triangles and quadrilaterals (chapter 12)

Types of triangles



Equilateral
3 equal sides, ∠s = 60°

Isosceles
2 equal sides and ∠s

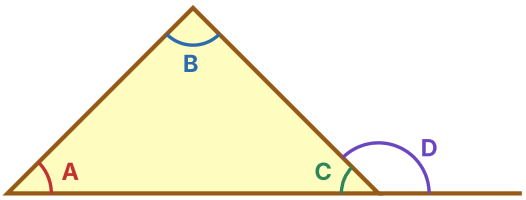
Scalene
no equal sides

Right-angled
one 90° angle

Tick marks show equal sides; a small square shows a right angle

Triangles can also be named by their angles: **acute-angled** (all angles less than 90°), **right-angled** (one 90°) and **obtuse-angled** (one angle more than 90°).

Angles of a triangle



A + B + C = 180° D = A + B

Interior angles and the exterior angle D

- The **interior angles** of a triangle add up to **180°** (sum of ∠s of △).
- The **exterior angle** of a triangle is equal to the **sum of the two opposite interior angles** (ext ∠ of △).
- In an **isosceles** triangle, the angles opposite the equal sides are equal (∠s opp equal sides).
- The longest side is opposite the biggest angle.

Example: A triangle has angles 2x, 3x and 85°. Find x.
2x + 3x + 85° = 180° (sum of ∠s of △) → 5x = 95° → **x = 19°**

Example: An isosceles triangle has a top angle of 40°. Find the two base angles.
The base angles are equal: (180° − 40°) ÷ 2 = **70°** each

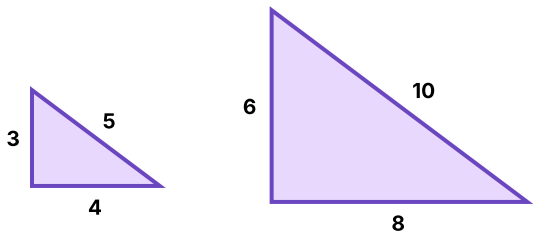
Congruent triangles (≡)

Congruent triangles are **exactly the same shape and size** – one fits perfectly on top of the other. Two triangles are congruent if one of these conditions is true:

Condition	What must be equal
SSS	All three sides
SAS	Two sides and the angle **between** them (included angle)
AAS	Two angles and a corresponding side
RHS	A right angle, the hypotenuse and one other side

AAA and SSA do **not** prove congruency. Write the letters in matching order: △ABC ≡ △DEF means A matches D, B matches E, C matches F.

Similar triangles (|||)



Every side of the big triangle is 2 times the small one

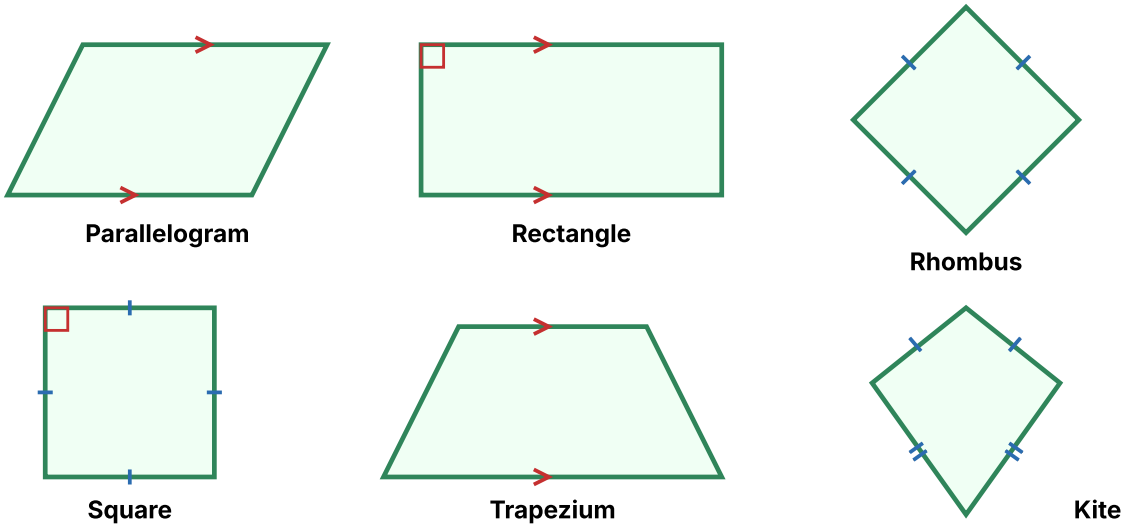
Similar triangles have the **same shape but a different size**. Two triangles are similar if:

- all their corresponding **angles are equal**, or
- all their corresponding **sides are in the same ratio** (proportion).

Example: △ABC ||| △DEF. AB = 4, DE = 8 and BC = 5. Find EF.
Ratio = DE ÷ AB = 8 ÷ 4 = 2 → EF = 5 × 2 = **10**

Quadrilaterals

A **quadrilateral** has **four sides**. Its interior angles add up to **360°**.



Arrows show parallel sides; tick marks show equal sides

Shape	Sides	Angles	Diagonals
Parallelogram	Both pairs of opposite sides parallel and equal	Opposite angles equal	Bisect each other (cut each other in half)
Rectangle	Opposite sides parallel and equal	All four angles 90°	Equal in length and bisect each other
Rhombus	All four sides equal; opposite sides parallel	Opposite angles equal	Bisect each other at 90°; bisect the angles
Square	All four sides equal; opposite sides parallel	All four angles 90°	Equal, bisect each other at 90°, bisect the angles
Trapezium	**One** pair of opposite sides parallel	–	–
Kite	Two pairs of **adjacent** sides equal	One pair of opposite angles equal	Perpendicular; the long one bisects the short one

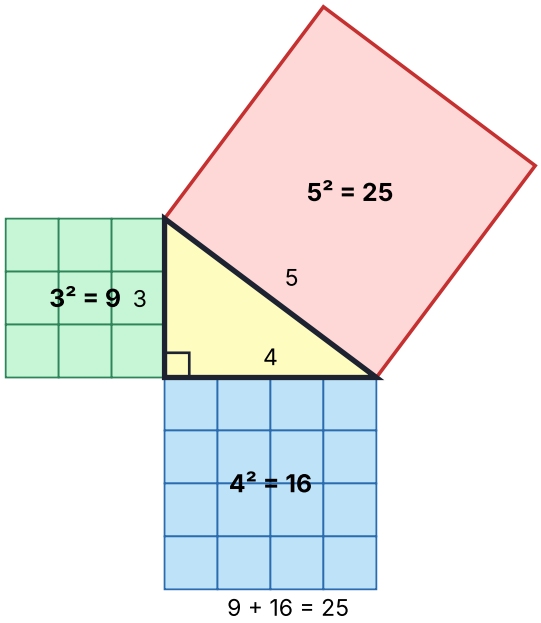
A square is a special rectangle **and** a special rhombus. A rectangle and a rhombus are special parallelograms.

Example: Three angles of a quadrilateral are 85°, 110° and 75°. Find the fourth angle.
360° − (85° + 110° + 75°) = 360° − 270° = **90°** (sum of ∠s of a quad)

The interior angles of **any polygon** add up to **180° × (n − 2)**, where n is the number of sides. A pentagon: 180° × 3 = 540°; a hexagon: 180° × 4 = 720°.

Paper 2 · 13. The Theorem of Pythagoras (chapter 13)

The Theorem of Pythagoras only works in a **right-angled triangle**. The longest side, opposite the right angle, is called the **hypotenuse**.



The square on the hypotenuse = the sum of the squares on the other two sides

> **Theorem of Pythagoras:** in a right-angled triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides.
>
> **$c^2 = a^2 + b^2$**
> (c = hypotenuse)

Use it to:

- find the **hypotenuse** (add the squares),
- find a **shorter side** (subtract the squares),
- **test** whether a triangle is right-angled.

Find the hypotenuse: the two shorter sides are 6 cm and 8 cm.
$c^2 = 6^2 + 8^2 = 36 + 64 = 100 \rightarrow c = \sqrt{100} =$ **10 cm**

Find a shorter side: the hypotenuse is 13 cm and one side is 5 cm.
$a^2 = 13^2 - 5^2 = 169 - 25 = 144 \rightarrow a = \sqrt{144} =$ **12 cm**

Is it right-angled? A triangle has sides 7, 24 and 25.
Longest side$^2 = 25^2 = 625$. Other two: $7^2 + 24^2 = 49 + 576 = 625$. They are equal, so **yes, it is right-angled** (converse of Pythagoras).

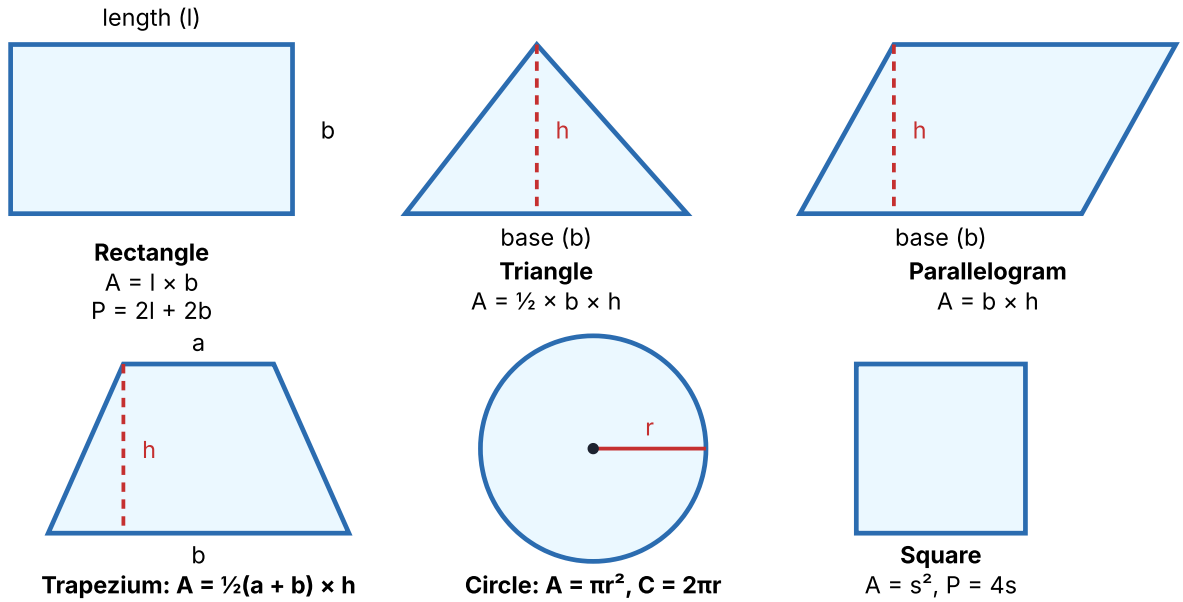
Word problem: a 5 m ladder leans against a wall. Its foot is 3 m from the wall. How high up the wall does it reach?
The ladder is the hypotenuse: $h^2 = 5^2 - 3^2 = 25 - 9 = 16 \rightarrow h =$ **4 m**

Answer that is not a whole number: sides 4 cm and 7 cm.
$c^2 = 16 + 49 = 65 \rightarrow c = \sqrt{65} =$ **8.06 cm** (round to 2 decimal places)

Pythagorean triples (whole-number right triangles): 3-4-5, 5-12-13, 8-15-17, 7-24-25 and their multiples (6-8-10, 9-12-15 …).

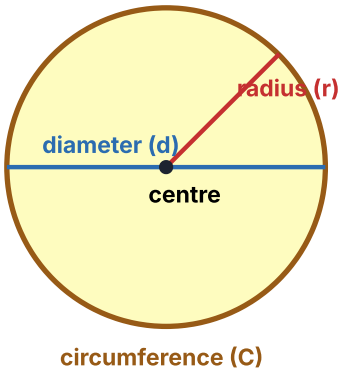
Paper 2 · 14. Perimeter and area (chapter 14)

Perimeter is the **distance around** a shape (units: mm, cm, m). **Area** is the **space inside** a flat shape (units: mm², cm², m²).



*Area formulas. The height (h, red dashes) is always **perpendicular** (at 90°) to the base.*

The circle



Parts of a circle

- **Radius (r):** from the centre to the edge.
- **Diameter (d):** right across through the centre. **d = 2r**.
- **Circumference (C):** the perimeter of a circle. **C = 2πr** (or **πd**).
- **Area: A = πr²**.
- **π ≈ 3.14** (or use the π key on your calculator).

Circle example: a circle has a diameter of 10 cm.
r = 10 ÷ 2 = 5 cm
Circumference: C = 2 × 3.14 × 5 = **31.4 cm**
Area: A = 3.14 × 5² = 3.14 × 25 = **78.5 cm²**

Triangle example: base 12 cm, perpendicular height 7 cm. A = ½ × 12 × 7 = **42 cm²**

Trapezium example: parallel sides 6 m and 10 m, height 4 m. A = ½ × (6 + 10) × 4 = ½ × 16 × 4 = **32 m²**

Combined shape: a rectangle 10 m × 6 m with a semicircle of diameter 6 m on one end.
Rectangle: 10 × 6 = 60 m²; semicircle: ½ × 3.14 × 3² = 14.13 m²; total = **74.13 m²**

Find a missing side: a rectangle has an area of 48 cm² and a length of 8 cm. Breadth = 48 ÷ 8 = **6 cm**

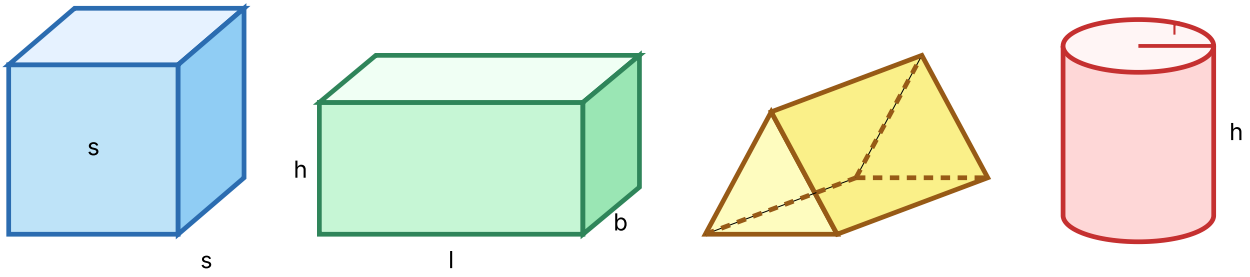
Converting units

Length	Area (square the conversion)
1 cm = 10 mm	1 cm² = 10 × 10 = **100 mm²**
1 m = 100 cm	1 m² = 100 × 100 = **10 000 cm²**
1 km = 1 000 m	1 km² = 1 000 000 m²

If all the lengths of a shape are multiplied by k, the **perimeter** is multiplied by k and the **area** by **k²**. Doubling the sides of a square makes its area 4 times bigger.

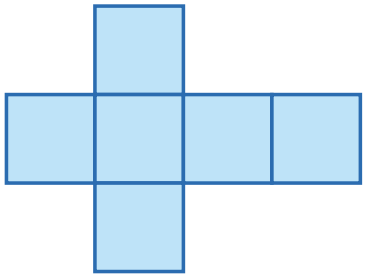
Paper 2 · 15. Volume and surface area (chapter 15)

Surface area is the **total area of all the faces** of a 3D object (units: cm², m²). **Volume** is the **space inside** a 3D object (units: cm³, m³).



The 3D objects you must know

Nets and surface area



6 equal squares → SA = 6s²

A net of a cube

A **net** is the flat shape you get when you unfold a 3D object. The surface area is the **area of the net**: add the areas of all the faces.

Object	Surface area
Cube	$6s^2$
Rectangular prism	$2(lb + lh + bh)$
Triangular prism	2 × area of triangle + area of the 3 rectangles
Cylinder	$2\pi r^2 + 2\pi rh$ (two circles + the curved side)

Volume

Volume of any prism = area of the base × height (the base is the shape that is the same all the way through).

Object	Volume
Cube	$s \times s \times s = s^3$
Rectangular prism	$l \times b \times h$
Triangular prism	(½ × base × height of the triangle) × length of the prism
Cylinder	$\pi r^2 \times h$

Rectangular prism 5 cm × 3 cm × 4 cm:
Volume = 5 × 3 × 4 = **60 cm³**
Surface area = 2(5×3 + 5×4 + 3×4) = 2(15 + 20 + 12) = 2 × 47 = **94 cm²**

Cube with sides of 4 cm: Volume = 4^3 = **64 cm³**; Surface area = 6 × 4^2 = 6 × 16 = **96 cm²**

Cylinder r = 7 cm, h = 10 cm (π = 3.14):
Volume = 3.14 × 7^2 × 10 = 3.14 × 49 × 10 = **1 538.6 cm³**
Surface area = 2 × 3.14 × 7^2 + 2 × 3.14 × 7 × 10 = 307.72 + 439.6 = **747.32 cm²**

Triangular prism: the triangle has a base of 6 cm and a height of 4 cm; the prism is 10 cm long.
Volume = (½ × 6 × 4) × 10 = 12 × 10 = **120 cm³**

Volume and capacity

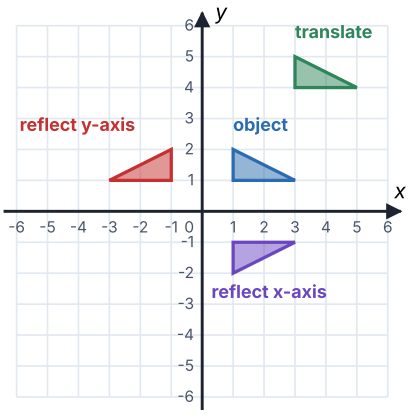
Volume	Capacity
1 cm^3	= 1 mℓ
1 000 cm^3	= 1 ℓ
1 m^3	= 1 000 ℓ = 1 kℓ

Example: a fish tank 50 cm × 30 cm × 40 cm holds 50 × 30 × 40 = 60 000 cm^3 = **60 ℓ** of water.

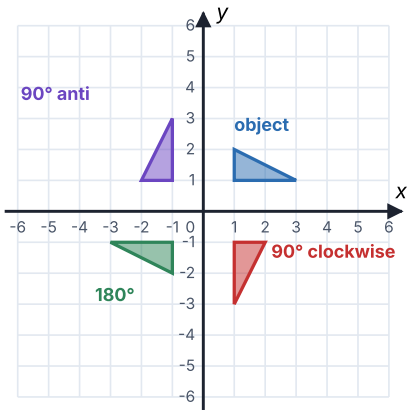
If all the lengths are multiplied by k, the surface area is multiplied by $\mathbf{k^2}$ and the volume by $\mathbf{k^3}$. Doubling the sides of a cube makes its volume **8 times** bigger.

Paper 2 · 16. Transformations (chapter 17)

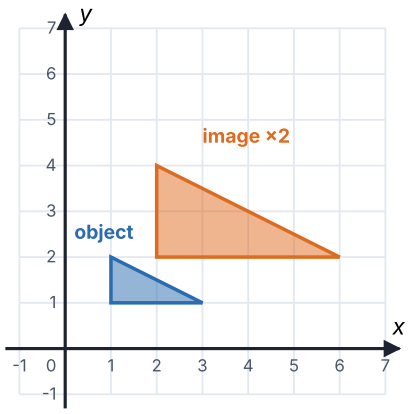
A **transformation** moves or changes a shape. The original shape is the **object**; the new shape is the **image**, labelled with a dash: A → A′.



Reflections and a translation (2 right, 3 up)



Rotations about the origin



Enlargement by a factor of 2 from the origin

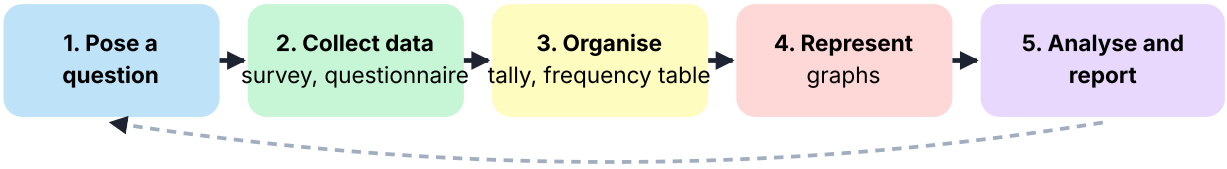
The blue **object** triangle has corners (1; 1), (3; 1) and (1; 2).

Transformation	What happens	Rule	Object (3; 1) becomes
Translation (slide)	Every point moves the same distance in the same direction	(x; y) → (x + a; y + b) 2 right, 3 up: (x + 2; y + 3)	(5; 4)
Reflection in the y-axis	Mirror image across the y-axis	(x; y) → (−x; y)	(−3; 1)
Reflection in the x-axis	Mirror image across the x-axis	(x; y) → (x; −y)	(3; −1)
Reflection in y = x	Swap x and y	(x; y) → (y; x)	(1; 3)
Rotation 90° clockwise about the origin	A quarter turn clockwise	(x; y) → (y; −x)	(1; −3)
Rotation 90° anticlockwise	A quarter turn anticlockwise	(x; y) → (−y; x)	(−1; 3)
Rotation 180°	A half turn	(x; y) → (−x; −y)	(−3; −1)
Enlargement by factor k from the origin	Shape gets bigger (k > 1) or smaller (0 < k < 1)	(x; y) → (kx; ky)	k = 2: (6; 2)

Same or different? Translations, reflections and rotations give an image that is **congruent** (same shape and size). An enlargement or reduction gives an image that is **similar** (same shape, different size). With a scale factor k, the area is multiplied by $\mathbf{k^2}$.

Paper 2 · 17. Data handling (chapter 19)

Data handling is collecting, organising, showing and interpreting information (data) to answer a question.



The data cycle

Collecting data

- **Population:** the whole group you want to know about. **Sample:** a smaller part of the population that you actually ask.
- A sample must be **representative** (like the whole population) and **not biased** (not favouring one answer). Asking only rugby players about the favourite sport of the school would be biased.
- **Categorical data:** words or groups (favourite colour). **Numerical data:** numbers – **discrete** (counted, e.g. number of siblings) or **continuous** (measured, e.g. height, mass).

Organising data

Tally and frequency table – marks: 25 learners' test marks (out of 10)

Mark	Tally	Frequency
5	𝍸	4
6	𝍸 |	6
7	𝍸 |||	8
8	𝍸	5
9	||	2
Total		**25**

Stem-and-leaf display – the stem is the tens digit, the leaves are the units: data 23, 25, 31, 34, 34, 38, 42, 47

Stem	Leaf
2	3 5
3	1 4 4 8
4	2 7

Measures of central tendency and spread

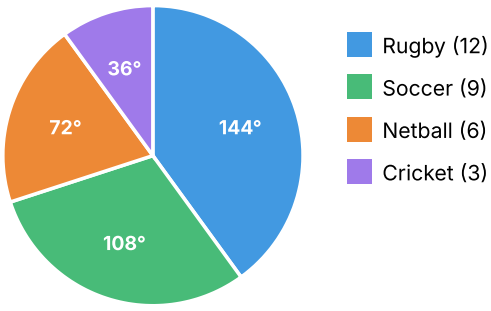
Data: **4; 7; 7; 9; 13**

Measure	What it is	How to find it	Answer
Mean (average)	The typical value	Add all values ÷ number of values	40 ÷ 5 = **8**
Median	The middle value	Put the data **in order** and find the middle. With an even number of values, add the two middle ones ÷ 2	**7**
Mode	The most common value	The value that appears most often (there can be more than one, or none)	**7**
Range	How spread out the data is	Highest value − lowest value	13 − 4 = **9**

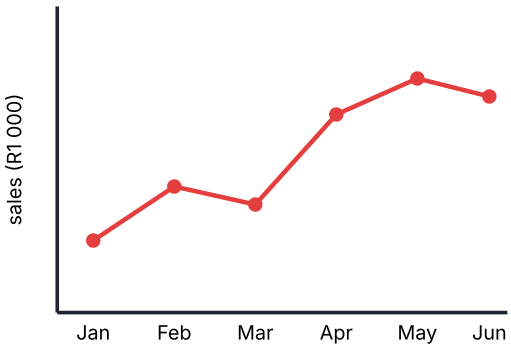
Median of an even set: 3; 5; 8; 10; 12; 15 → middle two are 8 and 10 → median = (8 + 10) ÷ 2 = **9**

An **outlier** is a value much bigger or smaller than the rest. It pulls the **mean** up or down a lot, so the **median** is a better "typical" value when there are outliers.

Graphs



Pie chart: favourite sport of 30 learners



Broken-line graph: change over time

Graph	Use it for	Important
Bar graph	Comparing categories (favourite sport, colours)	Bars are the same width with equal **gaps**
Double bar graph	Comparing two sets of data side by side (boys vs girls)	Use a key
Histogram	Grouped numerical data in intervals (marks 0–9, 10–19 ...)	Bars **touch** – no gaps
Pie chart	Parts of a whole (how a total is divided)	Angle of a sector = (frequency ÷ total) × 360°
Broken-line graph	Showing a change over time (temperature, sales)	Time goes on the horizontal axis
Scatter plot	Showing whether two variables are related (correlation)	Positive, negative or no correlation

Pie chart angles for 30 learners: Rugby 12 → (12 ÷ 30) × 360° = **144°**; Soccer 9 → **108°**; Netball 6 → **72°**; Cricket 3 → **36°**. Check: 144 + 108 + 72 + 36 = 360° ✓

Percentage from a pie chart: Rugby = 12 ÷ 30 × 100 = **40%** of the learners

Interpreting and reporting

- Describe what the graph shows: the **biggest** and **smallest** values, **trends** (going up or down), and any **outliers**.
- Use the mean, median, mode and range to support your conclusions.
- **Misleading graphs:** watch for a vertical axis that does not start at 0, uneven scales, missing labels or pictures of different sizes – these can make a small difference look big.

Definitions to learn by heart

These use the usual Grade 9 wording. If your teacher or textbook words a definition differently, learn their wording.

Term	Definition
Term	Parts of an expression separated by + or − signs.
Coefficient	The number in front of a variable.
Like terms	Terms with the same variables and the same exponents.
Factorise	Write an expression as a product of its factors.
Equation	A mathematical sentence with an equal sign; solving it means finding the value of the variable that makes it true.
Gradient	The steepness of a line: rise ÷ run.
y-intercept	The point where a graph cuts the y-axis ($x = 0$).
x-intercept	The point where a graph cuts the x-axis ($y = 0$).
Probability	How likely an event is to happen, from 0 (impossible) to 1 (certain).
Sample space	The list of all possible outcomes.
Relative frequency	The number of times an event happens ÷ the number of trials.
Acute / right / obtuse / straight / reflex angle	Less than 90° / 90° / between 90° and 180° / 180° / between 180° and 360°.
Complementary angles	Two angles that add up to 90°.
Supplementary angles	Two angles that add up to 180°.
Vertically opposite angles	The opposite angles formed when two lines cross; they are equal.
Transversal	A line that cuts across two or more other lines.
Corresponding angles	Angles in the same position at each line (F-shape); equal if the lines are parallel.
Alternate angles	Angles between the lines on opposite sides of the transversal (Z-shape); equal if the lines are parallel.
Co-interior angles	Angles between the lines on the same side of the transversal (U-shape); add up to 180° if the lines are parallel.
Perpendicular lines	Lines that meet at 90°.
Parallel lines	Lines that are always the same distance apart and never meet.
Congruent	Exactly the same shape and size.
Similar	The same shape but a different size; angles equal and sides in proportion.

Hypotenuse	The longest side of a right-angled triangle, opposite the right angle.
Theorem of Pythagoras	In a right-angled triangle, the square on the hypotenuse equals the sum of the squares on the other two sides.
Perimeter	The total distance around a shape.
Area	The amount of flat space inside a shape, in square units.
Circumference	The perimeter of a circle.
Radius / diameter	The distance from the centre to the edge of a circle / the distance across a circle through the centre (d = 2r).
Surface area	The total area of all the faces of a 3D object.
Volume	The amount of space inside a 3D object, in cubic units.
Net	A flat shape that folds up to make a 3D object.
Translation	Sliding a shape so every point moves the same distance in the same direction.
Reflection	Flipping a shape over a mirror line.
Rotation	Turning a shape around a fixed point.
Enlargement	Making a shape bigger or smaller by a scale factor.
Mean	The sum of all values ÷ the number of values.
Median	The middle value when the data is in order.
Mode	The value that occurs most often.
Range	The highest value − the lowest value.
Outlier	A value that is much bigger or smaller than the rest of the data.
Population / sample	The whole group being studied / a smaller part of it that is actually surveyed.